

Karan Bhagat

📞 214-257-4630 ✉ kbhagat0618@gmail.com [in/kbhagat04](https://www.linkedin.com/in/kbhagat04) github.com/kbhagat04 🌐 Personal Website

EDUCATION

Texas A&M University

Expected May 2026

Bachelor of Science in Computer Science, Minor in Business (GPA: 3.96 / 4.00)

College Station, TX

- **Relevant Coursework:** Programming Design and Concepts (C++), Discrete Mathematics, Data Structures and Algorithms, Linear Algebra, Computer Architecture, Software Security, Computer and Network Security

EXPERIENCE

Software Development Intern

May 2025 – August 2025

Arup

New York, NY

- Enhanced web applications by integrating Angular frontends with AWS Lambda/DynamoDB and Serverless Framework, and collaborated on database schema design to improve scalability and maintainability.
- Engineered large-scale data pipelines in Python (Pandas, RegEx, APIs) to parse 22GB+ energy datasets, analyze usage patterns, and automate data integration into PostgreSQL.
- Developed real-time monitoring systems for maritime safety using AIS/LiDAR ingestion, WebSocket/NMEA parsing, geospatial analysis, and SQL Server automation with alert workflows.

Software Engineering Intern

May 2024 – July 2024

Cuvasol Technologies

Dallas, TX

- Developed a resume screening application with Flask and Flutter, enabling automated mock interview preparation for clients while following Agile methodology.
- Implemented a React and MySQL document upload system to streamline resume submission and processing workflows.
- Extracted relevant skills and experience using LLMs, and generated technical interview questions with Retrieval-Augmented Generation (RAG) based on job descriptions.

Web Developer

August 2022 – July 2023

Paramount Technologies LLC

Dallas, TX

- Developed and styled company website using HTML, CSS, and JavaScript, featuring smooth scrolling functionality: <https://www.paramounttechllc.com/>.
- Implemented a user-friendly contact form with CSS styling and integrated email confirmation mechanism using PHP.
- Designed a distinctive logo and favicon for brand identity, ensuring seamless recognition across web browsers

PROJECTS

SHIELDS | Capstone project with L3Harris

- Engineered an automated vulnerability + compliance scanning pipeline using Python, Paramiko, and OpenSCAP, enabling secure remote execution of CIS compliance scans on Rocky Linux VMs and eliminating manual SSH operations.
- Developed end-to-end data workflow to run scans, retrieve XML results, convert to standardized JSON, and integrate with LLM-based remediation generation and Ansible Playbook, improving scan efficiency and accuracy by over 70%.

YouTune | Full stack web app

- Collaborated on the development of a short-form music streaming platform using Electron, React.js, and Django, with a Django REST API supporting user authentication and CRUD operations on playlists and songs: <https://github.com/kbhagat04/YouTune>
- Deployed APIs to Azure App Service with a PostgreSQL backend and tested endpoints via Postman to ensure reliability and scalability.
- Optimized search performance by implementing lazy loading, reducing initial response time by 80% and increasing daily search capacity by 5x through more efficient YouTube API usage.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, HTML5, CSS, Scheme, R, LaTeX, Swift

Technologies: React.js, Electron, Django, AWS, Azure, Pandas, REST APIs, SQL Server, Git, GitHub, Docker, JavaFX, MS Office, Adobe Suite, Google Cloud/Drive

Honors: Craig and Galen Brown Engineering Honors, Deans List, President's Endowed Scholarship, IB Diploma, National Merit Finalist